Development of an Automated Parametric Posture Solver for Complex Closed Kinematic Chains

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1 Introduction and Objectives

Recent collaborations between NASA and the European Space Agency (ESA) have highlighted the need for preliminary design tools capable of guaranteeing robust spacecraft control performance during early design phases [1]. ISAE-SUPAERO addressed this by developing the Satellite Dynamics Toolbox library (SDTlib) to model complex multi-body space flexible structures [2]. SDTlib relies on the Linear Fractional Transformation (LFT) paradigm to account for parametric uncertainties [3].

However, in closed kinematic chains, the number of degrees of freedom is less than the number of parameters required by SDTlib. This results in implicit constraints and because these constraints are typically non-rational, the constrained parameters cannot be directly represented as LFTs of the free parameters. This can be resolved by approximating the LFTs with rational functions, however, these approximations require the computation of the constrained parameters from the free parameters. Manually finding explicit expressions to perform this computation is often impossible. Therefore, the primary objective of this research is to address this by developing an algorithm that takes implicit constraints and free parameter values and then outputs the corresponding constrained parameter values with no intervention from the user. To verify and validate this tool's applicability, the solver will ultimately be tasked with evaluating complex mechanisms such as the hexapod found aboard ESA's Athena space telescope [4].

2 Methodology

To ensure the solver operates parametrically without *a priori* user intervention, the architecture was shifted away from traditional block-diagram interfaces to a custom text-based engine, employing the following workflow:

- **Abstract Mechanism Definition**: Mechanisms are defined using a standardized JSON data structure.
- **Automated Loop Discovery**: The solver parses the JSON file and translates it into a mathematical graph where bodies act as nodes and joints act as edges. This populates an adjacency matrix.
- **Cycle Extraction**: By feeding the adjacency matrix into MATLAB's `cyclebasis` function, the solver autonomously discovers all independent closed kinematic loops within the system.

3 Mathematical Formulation

Once the loops are discovered, the solver automatically detects whether the mechanism is two-dimensional or three-dimensional:

- **Planar Systems**: Two-dimensional kinematic loops are evaluated using a global angle approach. Compared to the more common local angle approach, the global angle approach decouples nonlinear trigonometric terms to improve computational efficiency. The solver minimizes the mechanism's x and y positions to enforce the loop closure constraints using the following generalized constraint equations:

$$\begin{cases} \sum_{i=1}^{n} (l_i + d_i) \cos \theta'_i = 0 \\ \sum_{i=1}^{n} (l_i + d_i) \sin \theta'_i = 0 \end{cases} \tag{1}$$

  where l_i is the link length, d_i is the prismatic joint/linear actuator length, and θ'_i is the global angle.

- **Spatial Systems**: Three-dimensional kinematic loops use the principal axis-angle method. This reduces the rotational residual from a 3×3 matrix to a 3×1 orientation error vector. The spatial loop closure constraints state that the total rotation around the loop must yield the identity matrix, I_3, and the cumulative translation must be zero, yielding six residuals to minimize per loop:

$$\begin{cases} R_{total} = \prod_{k=1}^{n} R(\hat{a}_k, \theta_k) = I_3 \\ r_{pos} = \sum_{i=1}^{n} R_{1:i} v_i = 0 \end{cases} \tag{2}$$

To drive the loop-closure residuals to zero, the vectors are fed into the Levenberg-Marquardt optimization algorithm. This approach dynamically resolves near-singular configurations where standard Newton-Raphson methods fail, ensuring robust convergence [5].

4 Results

The solver's mathematical precision and scalability were validated and verified across a series of test.

Baseline and Planar Tests: The solver was first verified against the four-bar linkage, acting as an initial benchmark. It achieved a maximum translational residual of 2.78×10^{-17} m as shown in Figure 1. This represented high numerical accuracy. Following the four-bar mechanism, the algorithm successfully parsed and solved the posture of increasingly complex, multi-loop planar systems such as the Jansen and Stephenson linkages without any user intervention.

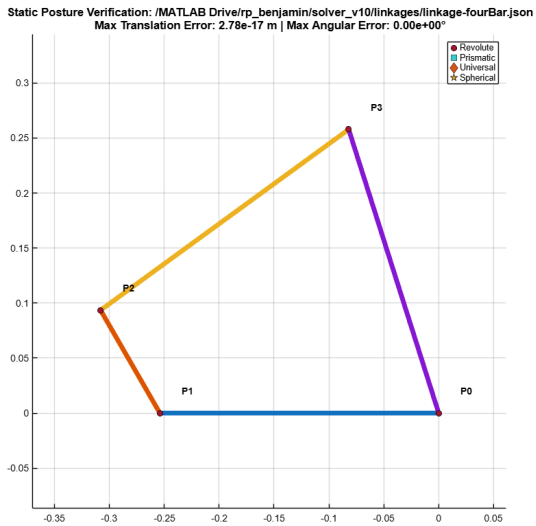

Figure 1: Solved four-bar mechanism posture.

Spatial Validation: To achieve the project's primary objective, the solver was tasked with evaluating ESA's Athena spacecraft's secondary mirror mechanism, the 6x3 hexapod [4].

Because of the hexapod's 6x3 configuration, specific topological workarounds were required. This configuration features coincident joints sharing the same physical space on its moving platform, imposing constraints impossible to handle with the solver. As such, kinematic equivalence had to be manually applied within the JSON definition. By artificially separating these shared connections into independent connections, the solver was able to bypass the previously limiting constraints and successfully generate the necessary constraint equations to make the system solvable.

The custom solver solved the hexapod's posture, achieving high accuracy with a maximum translation error of just 2.62×10^{-16} m. This is shown in Figure 2. Furthermore, a verification against established analytical inverse kinematics equations was performed. This showed near-identical linear actuator lengths down to an error on the order of magnitude of 10^{-16} m, highlighting the solver's robustness.

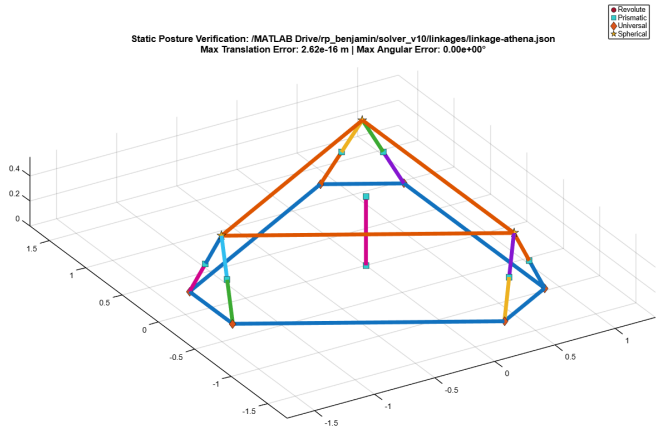

Figure 2: Solved hexapod mechanism posture.

5 Conclusion and Future Work

By successfully solving the posture of complex closed kinematic chains for their constrained parameters, the developed solver effectively bridges the gap between physical multi-body systems and the approximation of LFTs. The degree of novelty in this research lies in its automated graph-theory loop discovery and its planar and spatial mathematical formulation, which completely eliminate the need for lengthy and demanding manual loop closure equation derivations.

Future work should focus on automating singularity detection to computationally address redundant constraints like those found in the 6x3 ATHENA hexapod configuration in Section 4.

References

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[5] Tomomichi Sugihara. "Solvability-unconcerned inverse kinematics by the Levenberg–Marquardt method". In: *IEEE Transactions on Robotics* 27.5 (2011).